

Advanced (Carolina Reaper)

- Q1.** The absolute value of x is denoted by $|x|$. If $|x| = 5$, what are the possible values of x ?
- (A) 0 or 5
(B) -5 or 5
(C) 5 only
(D) 0 only
(E) -5 only
- Q2.** The distance between -3 and 4 on the number line is $|(-3) - 4| =$
- (A) 1
(B) 3
(C) 7
(D) 11
(E) 13
- Q3.** Tom has been saving money for a new bike and has $|-25| + 30$ dollars. How much money does Tom have in total?
- (A) 5 dollars
(B) 30 dollars
(C) 40 dollars
(D) 55 dollars
(E) 60 dollars
- Q4.** Samantha's favorite hiking trail is $|8 - 12| + 5$ kilometers long. What is the total length of the trail?
- (A) 1 km
(B) 5 km
(C) 9 km
(D) 13 km
(E) 17 km
- Q5.** Ashley is $|15 - 20|$ minutes late for her piano lesson. How many minutes late is Ashley?
- (A) 0 minutes
(B) 3 minutes
(C) 5 minutes
(D) 15 minutes
(E) 20 minutes
- Q6.** Ethan wants to buy a new video game that costs $|-30| + 10$ dollars. How much will Ethan pay for the video game?
- (A) 10 dollars
(B) 20 dollars
(C) 30 dollars
(D) 40 dollars
(E) 50 dollars
- Q7.** Jessica has $|12 - 8| \cdot 2$ pencils in her backpack. How many pencils does Jessica have?
- (A) 4 pencils
(B) 6 pencils
(C) 8 pencils
(D) 12 pencils
(E) 16 pencils
- Q8.** A bookshelf has $|5 - 9| + 2$ shelves. How many shelves does the bookshelf have?
- (A) 2 shelves
(B) 4 shelves
(C) 6 shelves
(D) 7 shelves
(E) 8 shelves

Open Ended Questions (FRQ)

- Q9.** If $|x + 2| = 7$, what are the possible values of x ? Show all your work and explain your reasoning.
- Q10.** A bakery is $|10 - 15|$ kilometers away from the city center. If it takes $|5 - 8|$ hours to get to the bakery by car, what is the average speed of the car in kilometers per hour? Show all your work and explain your reasoning.

Chef's Tips

- Emphasize the importance of understanding absolute value as distance from zero on the number line.
- Encourage students to use real-life scenarios to illustrate their understanding of absolute value.
- Remind students to check their work and explain their reasoning for free-response questions.